

useful metric to evaluate over time to determine if the trader or broker is skilled at market timing. But it does require a sufficient number of observations and a large data set.

It is worth noting that some B/Ds and vendors refer to the VWAP comparison as a cost rather than a gain/loss or performance indication. For those parties, a positive value indicates a higher cost (thus underperformance) and a negative value indicates a lower cost (thus better performance) and is the complete opposite of the meaning in the formula above. Unfortunately, representation of costs, P/L, or G/L as a metric is not consistent across industry participants and investors need to be aware of these differences.

Participation Weighted Price (PWP) Benchmark

Participation weighted price (PWP) is a variation of the VWAP analysis. It is intended to provide a comparison of the average execution price to the likely realized price had they participated with a specified percentage of volume during the duration of the order.

For example, if the PWP benchmark is a 20% POV rate and the investor transacted 100,000 shares in the market starting at 10 a.m. the PWP-20% benchmark price is computed as the volume weighted average price of the first 500,000 shares that traded in the market starting at 10 a.m. (the arrival time of the order). It is easy to see that if the investor transacted at a 20% POV rate their order would have been completed once 500,000 shares traded in the market since $0.20 \times 500,000 = 100,000$ shares. The number of shares in a PWP analysis is equal to the number of shares traded divided by the specified POV rate.

The PWP PnL metric is computed as follows:

$$PWP \text{ Shares} = \frac{\text{Shares Traded}}{\text{POV Rate}} \quad (3.15)$$

PWP Price = volume weighted price of the first PWP shares starting at the arrival time t_0

$$PWP \text{ PnL}_{bp} = -1 \cdot \frac{P_{avg} - PWP \text{ Price}}{PWP \text{ Price}} \cdot 10^4_{bp} \quad (3.16)$$

The PWP benchmark also has some inherent limitations similar to the VWAP metric. First, while PWP does provide insight into fair and reasonable prices during a specified time horizon it does not allow easy comparison across stocks or across days due to different stock price volatility and